In [1]:
```
import numpy as np
import pandas as pd
from sklearn.tree import DecisionTreeClassifier,plot_tree
from sklearn.preprocessing import LabelEncoder
import matplotlib.pyplot as plt
```

In [2]:
```
df=pd.read_csv('/home/backup/ml_datasets/OnlineRetail1.csv',encoding='cp1252')
print(df.head())
print(df.isnull().sum())
```

```
  InvoiceNo StockCode                          Description  Quantity  \
0    536365    85123A   WHITE HANGING HEART T-LIGHT HOLDER         6
1    536365     71053                  WHITE METAL LANTERN         6
2    536365    84406B       CREAM CUPID HEARTS COAT HANGER         8
3    536365    84029G  KNITTED UNION FLAG HOT WATER BOTTLE         6
4    536365    84029E       RED WOOLLY HOTTIE WHITE HEART.         6

        InvoiceDate  UnitPrice  CustomerID         Country
0  01-12-2010 08:26       2.55     17850.0  United Kingdom
1  01-12-2010 08:26       3.39     17850.0  United Kingdom
2  01-12-2010 08:26       2.75     17850.0  United Kingdom
3  01-12-2010 08:26       3.39     17850.0  United Kingdom
4  01-12-2010 08:26       3.39     17850.0  United Kingdom
InvoiceNo           0
StockCode           0
Description      1454
Quantity            0
InvoiceDate         0
UnitPrice           0
CustomerID     135080
Country             0
dtype: int64
```

In [3]:
```
df=df.dropna(subset=['CustomerID'])
df=df[(df['Quantity']>0)&(df['UnitPrice']>0)]
df['TotalSpend']=df['Quantity']*df['UnitPrice']
```

In [4]:
```
customer_df=df.groupby('CustomerID').agg({
    'Quantity':'sum',
    'UnitPrice':'mean',
    'TotalSpend':'sum',
    'Country':'first'
}).reset_index()
```

In [5]:
```
threshold=customer_df['TotalSpend'].median()
customer_df['HighValue']=(customer_df['TotalSpend']>threshold).astype(int)

le=LabelEncoder()
customer_df['Country']=le.fit_transform(customer_df['Country'])

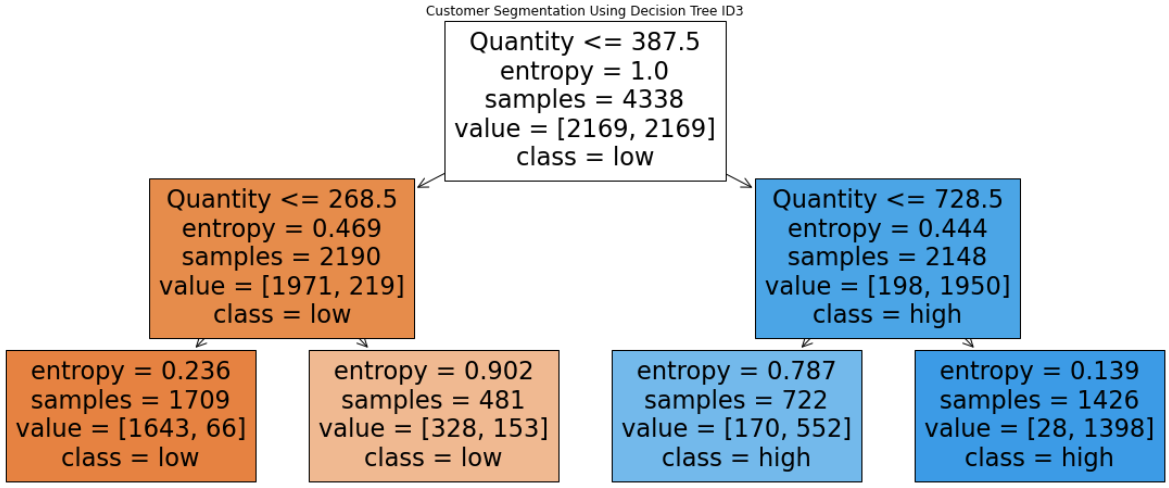
X=customer_df[['Quantity','UnitPrice','Country']]
y=customer_df['HighValue']
```

In [6]:
```
model=DecisionTreeClassifier(criterion='entropy',max_depth=2)
model.fit(X,y)
```

Out[6]:
```
DecisionTreeClassifier(criterion='entropy', max_depth=2)
```

In [7]:

```
plt.figure(figsize=(20,8))
plot_tree(model,feature_names=X.columns,class_names=['low','high'],filled=True)
plt.title("Customer Segmentation Using Decision Tree ID3")
plt.show()
```



In [8]:

```
importance=pd.Series(model.feature_importances_,index=X.columns)
print("feature importance :\n",importance.sort_values(ascending=False))
```

```
feature importance :
 Quantity     1.0
Country      0.0
UnitPrice    0.0
dtype: float64
```

In [ ]: